

AARYA SHIRSATH

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PROFILE

- Final-year B.Tech CSE student at VIT Bhopal (CGPA **8.6/10**) who has completed all required coursework and is **available to join full-time immediately**, targeting **SDE-1 backend roles** at product-driven startups.
- Builds and ships production-grade backend and AI systems in **Java, C++, and Python** — from REST APIs and ML pipelines to multi-agent LLM services — and **owns features end-to-end**: design, build, deploy, and debug in production.

EDUCATION

VIT Bhopal University — B.Tech, Computer Science & Engineering — CGPA **8.6/10**

2023 – 2027

- Coursework: Data Structures & Algorithms, Operating Systems (concurrency, multithreading), DBMS, OOP, System Design, Python Programming, Machine Learning, Advanced Java Programming.

TECHNICAL SKILLS

- Languages:** Java, C++, Python, SQL
- Backend & APIs:** FastAPI, REST API Design, Microservice Patterns, Pydantic, Uvicorn
- AI / ML:** LangChain, Multi-Agent Pipelines, RAG, Prompt Engineering, XGBoost, sklearn, LLM APIs (OpenAI, Anthropic, Groq)
- Cloud & DevOps:** AWS (EC2, S3, IAM, VPC), Docker, Git, GitHub, Postman, Linux
- Databases & Data:** PostgreSQL, MongoDB, SQLite, Pandas, NumPy

PROJECTS

MediAgent AI — AI Hospital Triage & Clinical Decision Support Platform

[GitHub](#) [Live Demo](#)

LangChain • Groq LLaMA-3.1 • Whisper v3 Turbo • Streamlit • SQLite • Plotly • OpenFDA API • FPDF

- Shipped a clinical triage platform (live on Streamlit Cloud) that takes patient symptoms by text or voice (Whisper-large-v3-turbo via Groq audio API) and routes cases across **13+ hospital departments** with explainable severity, **urgency (1–10)**, and confidence scores.
- Engineered a **three-agent LangChain pipeline** (Intake → Triage → Recommendation) on Groq LLaMA-3.1 behind **30+ deterministic red-flag guardrails** that force-escalate true emergencies and block unsupported 'Critical' calls — never trusting LLM output alone.
- Recently expanded the platform with three new modules: a **Lab Report Explainer** (OCR-based PDF/photo parameter extraction with trend charts), an **AI Health Chat assistant** (RAG-style Q&A via TF-IDF retrieval with clinical synonym expansion), and a **Pre-Appointment Prep Generator** (auto-generated doctor questions from recent case/lab history).

FraudShield AI — Real-Time Financial Fraud Detection System

[GitHub](#) [Live Demo](#)

Python • XGBoost • FastAPI • GPT-3.5 • Scikit-learn • HTML/CSS/JS

- Built a real-time fraud-scoring REST service (FastAPI, **sub-100 ms inference latency**) that flags suspicious transactions inside the payment window and returns a GPT-3.5-generated plain-language risk explanation for compliance review.
- Engineered **14 statistical features** (balance deltas, account-drain flags, transaction velocity) and trained an XGBoost classifier on **6.3M+ transactions**, tuned to **99% recall at 80% precision** on heavily imbalanced fraud data.

EduGrade-AI — AI-Powered Virtual Classroom & Assignment Evaluation Platform

[GitHub](#) [Live Demo](#)

Flask • SQLAlchemy • Google Document AI • Gemini • Bootstrap

- Built a Google Classroom-style platform (Flask, SQLAlchemy) that runs submitted PDFs through **Google Document AI** for OCR/layout extraction, then **Gemini** for LLM-based auto-grading with written feedback benchmarked against a teacher's ideal answer.
- Engineered a **two-layer plagiarism detector** — text-similarity scoring (sequence matching, cosine, trigram overlap) that escalates to a bounding-box-based handwriting comparison above a **90% similarity threshold**, with role-based access enforced per class enrollment.

INTERNSHIPS

MPOnline Internship (Under Govt. Of Madhya Pradesh)

May – July 2026

AI/ML Internship

[GitHub](#) [CNN Projects](#)

- Worked across the **ML lifecycle** — data preprocessing, model development, and evaluation — and built **3 CNN classifiers** (TensorFlow) for CIFAR-10 classification, LFW face recognition, and MRI brain-tumor classification, reaching **89% accuracy**.

SDE Internship (Advanced SWE & Development)

[GitHub](#)

- Designed and built **TaskBoard**, a full-stack Flask + MySQL task management system with a **bcrypt-secured login** and a REST API (GET/POST /api/tasks) letting admins log, track, and mark employee tasks complete on a live dashboard.

CERTIFICATIONS & ACHIEVEMENTS

- Top 10% globally (90th percentile)** — TCS CodeVita Season 13 competitive programming contest. [Certificate](#)
- HackerRank Certified Software Engineer** — cleared the role-based certification for the Software Engineer track. [Certificate](#)
- AWS training certification (Intellipaat)** — EC2, S3, IAM, VPC, cloud deployment and networking. [Certificate](#)
- ServiceNow Virtual Internship** — ServiceNow University × AICTE × SmartBridge: completed an **AICTE-affiliated internship** covering **ServiceNow administration, Flow Designer workflow automation, Agentic AI**, and the **Automated Test Framework (ATF)**; preparing for the **Certified System Administrator (CSA)** exam. [Certificate](#)
- Co-authored research paper “**Edge Computing for Intelligent Systems**” on distributed AI architectures for low-latency systems. [Paper](#)
- Additional coursework: **Coursera** (Computer Networking — TCP/IP, DNS, HTTP), **Vityarthi** (Java, Python). [Certificate](#)

LEADERSHIP & EXTRACURRICULAR

- Technical Core Member** — **IEEE Student Branch, VIT Bhopal** (2023–Present): organized ML and open-source workshops for **100+ members**; coordinated inter-college hackathon logistics end-to-end.
- Competitive programmer: completed the **50-day #DrGViswanathanChallenge** (DSA in Java, LeetCode); maintains public GitHub repos with documented architecture decisions. [GitHub](#)